

# LABORATORY OBSERVATION

**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Experiment No.:** 1

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**Roll No.:** A24126552067

**Date:** 31-08-2026

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## 1. TITLE

Design and Implementation of College Information Portal Using HTML5

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## 2. AIM

To develop a structured, semantic HTML5 web page for the ANITS College Information Portal that incorporates text formatting, department lists, student detail tables, interactive registration forms, and multimedia audio/video players.

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## 3. OBJECTIVE

The objectives of this experiment are:

- To understand core HTML5 semantic tags and document layout structure.
  - To implement formatted text, mathematical/chemical formulas using sup and sub tags.
  - To create structured tabular data using table, tr, th, and td elements.
  - To design an interactive student registration form with various HTML5 input types.
  - To embed and control multimedia audio and video players.
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## 4. THEORY

HTML5 is the standard markup language for creating modern web pages. It introduces semantic elements such as <header>, <nav>, <section>, <article>, and <footer> to structure content meaningfully.

Forms in HTML5 provide rich input controls including text, email, password, radio buttons, checkboxes, dropdown select, color pickers, and date pickers for client-side data capture.

HTML5 also provides native <audio> and <video> elements with browser controls for seamless media playback without third-party plugins.

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## 5. ALGORITHM / PROCEDURE

1. Create an HTML5 document and configure the title as 'College Information Portal'.
  2. Add a header section with institution title and introductory description.
  3. Create a navigation bar with hyperlink anchors to sections and external portals.
  4. Format about text using bold, italic, underline, superscript, and subscript tags.
  5. Create ordered list for departments and unordered list for campus facilities.
  6. Construct a student details table with borders, column headers, and records.
  7. Build a comprehensive student registration form with text, email, password, branch select, date, color, and range inputs.
  8. Embed audio and video tags with controls attribute.
  9. Open in web browser and verify all elements and form controls.
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## 6. PROGRAM

**Source File:** task1.html

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<!DOCTYPE html>
<html lang="en">
<!-- head is for the head of the html-->
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<!-- and we give body to the website or doc -->
<body>
  <div style="background-color: azure;">
    <header>
      <table border="1">
        <tr>
          <td>
            <nav>
              <a href="https://anits.org/" target="_blank"><br> ANITS Website </a>
|
              <a href="https://www.google.com" target="_blank"><br> Google Search
</a>
            </nav>
          </td>
        </tr>
      </table>
    </header>
    <H1>
    <!-- THE STRONG WE USED CAUSE ITSTHE MAIN HEADING AND WE WANT IN BOLD-->
    <strong>ANITS College Information Portal</strong>
    </H1>
    <!-- Adding the requested college logo -->
    
    <!-- br is for new line -->
    <br>
    <hr>
    <!-- so for the net features not t shwo like iots an doc we are giving the horizntal line
-->
  </div>

  <div style="background-color: rgb(206, 241, 230);">
    <section>
      <h2>
        <!-- THE B WE USED CAUSE ITSTHE 2ND MAIN HEADING AND WE WANT IN BOLD-->
        <b>About Our College</b>
      </h2>
      <p>
        <I>Welcome to ANITS. We focus on <U>excellence</U> in education.</I><br>
        here we can learn everything about studys and create new things new formulas like
        E = mc<SUP>2</SUP>
        and also formulas Chemical Formula: H<SUB>2</SUB>O
      </p>
    </section>
    <hr>
  </div>

```

```

<div style="background-color: aliceblue;">
  <article>
    <H1>Programs and Facilities</H1>
    <P>
      Here are the top programs we offer (Ordered List):
      <ol>
        <li><b>Computer Science Engineering</b></li>
        <li><b>Information Technology</b></li>
        <li><b>Electronics and Communication</b></li>
      </ol>
      And our facilities (Unordered List):
      <ul>
        <li>Library</li>
        <li>Advanced Labs</li>
        <li>Sports Complex</li>
      </ul>
    </P>
  </article>
  <hr>
</div>

<div style="background-color: antiquewhite;">
  <h2>
    Student Details Table
    <!-- table creating for the names of students -->
  </h2>
</div>

<div style="background-color: peachpuff;">
  <p>
    <table border="2">
      <tr>
        <th>Roll No</th>
        <th>Name</th>
        <th>Branch</th>
        <th>Email</th>
      </tr>
      <tr>
        <td>67</td>
        <td>A.Rohit</td>
        <td>CSE (AI & ML)</td>
        <td>rohit2906a@example.com</td>
      </tr>
    </table>
  </p>
  <br>

  <h3>CAMPUS</h3>
  <p>the below pic is our campus</p>
  <!-- A relevant campus image -->
  <img src=campus.png alt="Campus Building">
  <hr>
</div>

```

```

<div style="background-color: azure;">
  <section>
    <h2>Student Registration Form</h2>
    <!-- form with new HTML5 inputs -->
    <form action="#">
      Name: <input type="text" name="name" value="A.Rohit" required><br><br>
      Email: <input type="email" name="email" required><br><br>
      Password: <input type="password" name="password"><br><br>
      DOB: <input type="date" name="dob"><br><br>
      Age: <input type="number" name="age"><br><br>
      Favorite Color: <input type="color" name="color"><br><br>
      Skill known: <input type="range" name="skill" min="1" max="10"><br><br>

      Gender:
      <input type="radio" name="gender" value="male"> Male
      <input type="radio" name="gender" value="female"> Female<br><br>

      Hobbies:
      <input type="checkbox" name="hobbies" value="coding"> Coding
      <input type="checkbox" name="hobbies" value="gaming"> Gaming<br><br>

      Branch:
      <select name="branch">
        <option value="cse">CSE</option>
        <option value="ece">ECE</option>
        <option value="mech">MECH</option>
      </select><br><br>

      Comments:<br>
      <textarea name="comments" rows="3" cols="30"></textarea><br><br>

      <input type="submit" value="Submit">
      <input type="reset" value="Clear">
    </form>
  </section>
  <hr>
</div>

<div style="background-color: rgb(206, 241, 230);">
  <section>
    <h2>Media Elements</h2>
    <!-- Audio and Video tags -->
    <audio controls><source src="audio.mp3" type="audio/mpeg"></audio><br><br>
    <video controls width="250"><source src="video.mp4" type="video/mp4"></video>
  </section>
  <hr>

  <footer>
    <p>&copy; 2026 Developed by A.Rohit</p>
  </footer>
</div>

```

```
</body>
</html>
```

7. CODE EXPLANATION

| Code / Syntax                      | Explanation                                                               |
|------------------------------------|---------------------------------------------------------------------------|
| <header>, <nav>, <footer>          | Semantic HTML5 layout container elements.                                 |
| <sup>, <sub>                       | Displays text as superscript (X²) or subscript (H₂O).                     |
| <ol>, <ul>, <li>                   | Creates ordered (numbered) and unordered (bulleted) lists.                |
| <table>, <tr>, <th>, <td>          | Structures tabular data into rows, header cells, and standard data cells. |
| <input type='radio'>               | Radio button group allowing only one gender selection.                    |
| <select>, <option>                 | Drop-down select menu for branch selection.                               |
| <audio controls>, <video controls> | Native media playback controls for audio and video files.                 |

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

| Test Case | Input / Action   | Expected Result             | Actual Result                             | Status |
|-----------|------------------|-----------------------------|-------------------------------------------|--------|
| TC-01     | Page Load        | Open index.html in browser  | Header, nav, and sections render cleanly  | Pass   |
| TC-02     | Lists & Formulas | Verify lists and formulas   | Ordered/unordered lists display properly  | Pass   |
| TC-03     | Table Rendering  | Check student details table | Table renders columns with row borders    | Pass   |
| TC-04     | Form Inputs      | Fill form & click Submit    | Inputs accept data, radio selection works | Pass   |
| TC-05     | Media Playback   | Click play on audio/video   | Media plays with default browser controls | Pass   |

## 9. OUTPUT

### Output 1

[ANITS Website](#)  
[Google Search](#)

## ANITS College Information Portal



### About Our College

Welcome to ANITS. We focus on *excellence* in education.  
here we can learn everything about studys and create new things new formulas like  $E = mc^2$  and also formulas Chemical Formula:  $H_2O$

### Programs and Facilities

Here are the top programs we offer (Ordered List):

1. **Computer Science Engineering**
2. **Information Technology**
3. **Electronics and Communication**

And our facilities (Unordered List):

- **Library**

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## 10. RESULT

Thus, the College Information Portal was successfully implemented using HTML5, incorporating text formatting, lists, tables, registration form inputs, and multimedia elements, and verified across all test cases.